

Exploratory Data Analysis Summary: S&P500

Daily Returns | covers all tickers across January 2010 to December 2024

Data Source: Yahoo Finance API

Variables analyzed: percentage returns based on Adjusted Close, across all tickers.

Visualization Methods: Histograms per ticker, generated via scripts found on [GitHub](#).

1. Key Distribution Insight

Bell-shaped distributions centered around 0% for most tickers. With approximately 68% daily returns mostly within $\pm 2\%$.

Variable volatility across tickers:

Narrow curves characterized by tighter cluster around 0 indicate lower volatility

- Utilities sector
- Consumer Staples
- Real Estate

Wider curves characterized by spread-out histograms indicate higher volatility

- Information Technology Sector
- Biotechnology
- Energy

Occasional fat tails observed; few tickers exhibit more frequent large swings ($> \pm 5\%$)

- Eg. GME

2. Volatility & Distribution Notes

Standard deviation of daily returns ranges:

- Low volatility: $\leq 0.8\%$ (stable tickers)
- High volatility: $> 2.5\%$ (volatile ones).

Subtle positive skews in some tickers, suggesting small upward bias in daily performance.

3. Outliers & Anomalies

Outlier returns ($> \pm 5\%$) often linked to:

- Earnings Surprises
- Macro Shocks
- Sector-specific news

Histograms help flag tickers that experience frequent outliers.

4. Missing Data & Gaps

- Automated scripts confirmed no missing dates in cleaned data.

During the initial data collection phase, two S&P 500 tickers, BRK.B and BF.B, could not be processed due to formatting conflicts with their dot notation. These were flagged as unusable after resulting in empty files and have been excluded from modelling and analysis for now.

5. Modeling Implications

- Volatility differences across ticker groups suggest using volatility-specific features or grouping tickers by volatility during modeling.
- Daily returns are roughly normally distributed, regression models like Linear Regression or XGBoost can work well for predicting return magnitude.

6. Recommended Feature Engineering

- Volatility Measures: rolling standard deviation.
- Technical Indicators:
 - Trend: EMA (10) + MACD
 - Momentum: ROC (Rate of Change)
 - Volatility: Bollinger Bands
 - Volume: Volume % Change
- Volume Metrics: daily volume changes or spike indicators.